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Semiconductor device, inverter circuit, driving device, vehicle, and elevator

Abstract

A semiconductor device according to an embodiment includes a semiconductor chip having a transistor region and a diode region, and a conductor. The semiconductor chip includes a first electrode, a second electrode, a silicon carbide layer between the first electrode and the second electrode, and a gate electrode. The first electrode includes a first region in the transistor region and a second region in the diode region. A first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-139053, filed on Sep. 1, 2022, the entire contents of which are incorporated

herein by reference.

FIELD

(2) Embodiments described herein relate generally to a semiconductor device, an inverter circuit, a driving device, a vehicle, and an elevator.

BACKGROUND

(3) Silicon carbide (SiC) is expected as a material for a next-generation semiconductor device. As compared with silicon, silicon carbide has excellent physical properties such as a band gap of three times, a breakdown field strength of about ten times, and a thermal conductivity of about three times. By utilizing this characteristic, for example, a metal oxide semiconductor field effect transistor (MOSFET) capable of operating at a high breakdown voltage, a low loss, and a high temperature can be realized.

(4) A vertical MOSFET using silicon carbide includes a pn junction diode as a built-in diode. For example, the MOSFET is used as a switching element connected to an inductive load. In this case, even when the MOSFET is turned off, it is possible to allow a reflux current to flow using the built-in diode.

(5) However, when a reflux current is caused to flow using a body diode, there is a problem in that a stacking fault grows in a silicon carbide layer due to recombination energy of carriers, and on-resistance of the MOSFET increases. An increase in on-resistance of the MOSFET causes a decrease in reliability of the MOSFET. For example, by providing a Schottky Barrier Diode (SBD) that performs unipolar operation as a built-in diode in the MOSFET, it is possible to suppress growth of the stacking fault in the silicon carbide layer. The reliability of the MOSFET is improved by providing the SBD as a built-in diode in the MOSFET.

(6) A large surge current may flow into the MOSFET instantaneously beyond the steady state. When a large surge current flows, a large surge voltage is applied to generate heat, and the MOSFET is broken. The maximum allowable peak current value (I.sub.FSM) of the surge current allowed in the MOSFET is referred to as a surge current tolerance. In the MOSFET having the SBD provided therein, it is desired to improve a surge current tolerance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**A and **1**B are schematic diagrams of a semiconductor device according to a first embodiment;

(2) FIGS. **2**A and **2**B are schematic top views of the semiconductor device according to the first embodiment;

(3) FIG. **3** is a schematic cross-sectional view of the semiconductor device according to the first embodiment;

(4) FIG. **4** is a schematic cross-sectional view of the semiconductor device according to the first embodiment;

(5) FIGS. **5**A and **5**B are schematic cross-sectional views of the semiconductor device according to the first embodiment;

(6) FIG. **6** is a schematic top view of the semiconductor device according to the first embodiment;

(7) FIG. **7** is a schematic top view of the semiconductor device according to the first embodiment;

(8) FIGS. **8**A and **8**B are schematic top views of a semiconductor device according to a first comparative example;

(9) FIG. **9** is a schematic cross-sectional view of the semiconductor device according to the first comparative example;

(10) FIG. **10** is an equivalent circuit diagram of the semiconductor device according to the first comparative example;

(11) FIGS. **11**A and **11**B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment;
(12) FIG. **12** is a schematic cross-sectional view of a semiconductor device according to a second comparative example;
(13) FIG. **13** is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment;
(14) FIGS. **14**A and **14**B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment;
(15) FIG. **15** is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment;
(16) FIGS. **16**A and **16**B are schematic top views of a semiconductor device according to a third comparative example;
(17) FIG. **17** is a schematic top view of a semiconductor device according to a third comparative example;
(18) FIGS. **18**A and **18**B are explanatory diagrams of a problem of the semiconductor device according to the third comparative example;
(19) FIGS. **19**A and **19**B are schematic top views of a semiconductor device according to a second embodiment;
(20) FIG. **20** is a schematic top view of the semiconductor device according to the second embodiment;
(21) FIG. **21** is a schematic diagram of a driving device according to a third embodiment;
(22) FIG. **22** is a schematic diagram of a vehicle according to a fourth embodiment;
(23) FIG. **23** is a schematic diagram of a vehicle according to a fifth embodiment; and
(24) FIG. **24** is an elevator according to a sixth embodiment.

DETAILED DESCRIPTION

(25) A semiconductor device according to an embodiment includes: a semiconductor chip including a plurality of transistor regions and at least one diode region, wherein the transistor regions include a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane, a first electrode in contact with the first portions, the second silicon carbide region, and the third silicon carbide region, a second electrode in contact with the second plane, a gate electrode facing the second silicon carbide region, and a gate insulating layer provided between the gate electrode and the second silicon carbide region, wherein the at least one diode region includes the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane and a fourth silicon carbide region of p-type provided between the first silicon carbide region and the first plane, the first electrode in contact with the second portions and the fourth silicon carbide region, and the second electrode, wherein an occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per the unit area of the second silicon carbide region projected onto the first plane, and wherein a first diode region, which is one of the at least one diode region, is provided between a first transistor region, which is one of the transistor regions, and a second transistor region, which is one of the transistor regions provided in a first direction with respect to the first transistor region; and a conductor having one end in contact with the first electrode and applying a voltage to the first electrode, wherein the first electrode includes a first region in the transistor regions and a second region in the at least one diode region, and wherein a first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.
(26) Hereinafter, embodiments of the present disclosure will be described with reference to the

drawings. In the following description, the same or similar members and the like are denoted by the same reference numerals, and the description of the members and the like once described may be appropriately omitted.
(27) In the following description, when notations of n.sup.+, n, and n.sup.− and p.sup.+, p, and p.sup.− are used, the above notations represent relative levels of impurity concentration in each conductive type. That is, n.sup.+ indicates that the n-type impurity concentration is relatively higher than n, and n-indicates that the n-type impurity concentration is relatively lower than n. In addition, p.sup.+ indicates that the p-type impurity concentration is relatively higher than p, and p.sup.− indicates that the p-type impurity concentration is relatively lower than p. In addition, an n.sup.+ type and an n.sup.− type may be simply referred to as an n type, and a p.sup.+ type and a p.sup.− type may be simply referred to as a p type.
(28) The impurity concentration can be measured by, for example, secondary ion mass spectrometry (SIMS). Further, the relative level of the impurity concentration can also be determined from the level of carrier concentration obtained by, for example, scanning capacitance microscopy (SCM). In addition, distances such as a depth and a thickness of an impurity region can be obtained by, for example, SIMS. Furthermore, distances such as a depth, a thickness, a width, and an interval of the impurity region can be obtained from, for example, a composite image of an SCM image and an atomic force microscope (AFM) image.
(29) In the present specification, impurity concentration of a semiconductor region means the maximum impurity concentration of the semiconductor region unless otherwise stated.
First Embodiment
(30) A semiconductor device according to a first embodiment includes a semiconductor chip including a plurality of transistor regions and at least one diode region. The plurality of transistor regions include a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane, a first electrode in contact with the plurality of first portions, the second silicon carbide region, and the third silicon carbide region, a second electrode in contact with the second plane, a gate electrode facing the second silicon carbide region, and a gate insulating layer provided between the gate electrode and the second silicon carbide region. The at least one diode region includes the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane, and a fourth silicon carbide region of p-type provided between the first silicon carbide region and the first plane, the first electrode in contact with the plurality of second portions and the fourth silicon carbide region, and the second electrode. An occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per unit area of the second silicon carbide region projected onto the first plane. A first diode region, which is one of the at least one diode region, is provided between a first transistor region, which is one of the plurality of transistor regions, and a second transistor region, which is one of the plurality of transistor regions provided in a first direction with respect to the first transistor region. The semiconductor device according to the first embodiment includes a conductor having one end in contact with the first electrode and applying a voltage to the first electrode. The first electrode includes a first region in the plurality of transistor regions and a second region in the at least one diode region. A first contact area between the conductor and the first region is larger than a second contact area between the conductor and the second region.
(31) The semiconductor device according to the first embodiment is a discrete device **1000**. In the case of the discrete device **1000**, the discrete device **1000** has one MOSFET **100** mounted thereon with a sealing resin. The MOSFET **100** is an example of a semiconductor chip.
(32) The MOSFET **100** included in the discrete device **1000** is a vertical MOSFET of a planar gate

type using silicon carbide. The MOSFET **100** is, for example, a double implantation MOSFET (DIMOSFET) in which a body region and a source region are formed by ion implantation. The MOSFET **100** is a MOSFET including a Schottky barrier diode (SBD) as a built-in diode. The MOSFET **100** is a vertical MOSFET of an n-channel type using electrons as carriers.
(33) FIGS. **1**A and **1**B are schematic diagrams of the semiconductor device according to the first embodiment. FIG. **1**A is a top view of the discrete device **1000**. FIG. **1**B is a cross-sectional view of the discrete device **1000**, taken along line K-K′ in FIG. **1**A.
(34) The discrete device **1000** includes the MOSFET **100**, a first bonding wire **110***a* (conductor), a second bonding wire **110***b*, a third bonding wire **110***c*, a metal bed **120** (first metal layer), a first metal lead **130***a* (second metal layer), a second metal lead **130***b*, a third metal lead **130***c*, and a sealing resin **145**.
(35) The MOSFET **100** includes a silicon carbide layer **10**, a source electrode **12** (first electrode), a drain electrode **14** (second electrode), a gate electrode pad **22**, and a gate wiring **24**.
(36) The MOSFET **100** is disposed on the metal bed **120**. The MOSFET **100** is connected to the metal bed **120**.
(37) The metal bed **120** faces the drain electrode **14** of the MOSFET **100**. The metal bed **120** is an example of a first metal layer. The metal bed **120** is in contact with the drain electrode **14**. The metal bed **120** is electrically connected to the drain electrode **14**.
(38) The metal bed **120** is made of metal. The metal bed **120** is, for example, a copper-based alloy or an iron-nickel alloy.
(39) The first metal lead **130***a* extends in the first direction. The first metal lead **130***a* is an example of a second metal layer.
(40) The first metal lead **130***a* is an electrode terminal. The first metal lead **130***a* has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**.
(41) The first metal lead **130***a* is metal. The first metal lead **130***a* is, for example, a copper-based alloy or an iron-nickel alloy.
(42) The second metal lead **130***b* extends in the first direction. The second metal lead **130***b* is an electrode terminal. The second metal lead **130***b* has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**.
(43) The second metal lead **130***b* is metal. The second metal lead **130***b* is, for example, a copper-based alloy or an iron-nickel alloy.
(44) The third metal lead **130***c* extends in the first direction. The third metal lead **130***c* is an electrode terminal. The third metal lead **130***c* has a function of applying a voltage to the semiconductor chip **100** from the outside of the discrete device **1000**. The third metal lead **130***c* is connected to the metal bed **120**.
(45) The third metal lead **130***c* is metal. The third metal lead **130***c* is, for example, a copper-based alloy or an iron-nickel alloy.
(46) The first bonding wire **110***a* has one end connected to the source electrode **12** and the other end connected to the first metal lead **130***a*. The first bonding wire **110***a* is an example of a conductor.
(47) The first bonding wire **110***a* electrically connects the source electrode **12** to the first metal lead **130***a*. The first bonding wire **110***a* has a function of applying a voltage applied to the first metal lead **130***a* to the source electrode **12**.
(48) The first bonding wire **110***a* is metal. The first bonding wire **110***a* is, for example, aluminum, copper, or gold.
(49) The second bonding wire **110***b* has one end connected to the source electrode **12** and the other end connected to the first metal lead **130***a*. The second bonding wire **110***b* electrically connects the source electrode **12** to the first metal lead **130***a*. The second bonding wire **110***b* has a function of applying a voltage applied to the first metal lead **130***a* to the source electrode **12**.
(50) The second bonding wire **110***b* is metal. The second bonding wire **110***b* is, for example,

aluminum, copper, or gold.
(51) The third bonding wire **110***c* has one end connected to the gate electrode pad **22** and the other end connected to the second metal lead **130***b*. The third bonding wire **110***c* electrically connects the gate electrode pad **22** to the second metal lead **130***b*. The third bonding wire **110***c* has a function of applying a voltage applied to the second metal lead **130***b* to the gate electrode pad **22**.
(52) The third bonding wire **110***c* is metal. The third bonding wire **110***c* is, for example, aluminum, copper, or gold.
(53) The sealing resin **145** covers the MOSFET **100**, the first bonding wire **110***a*, the second bonding wire **110***b*, the third bonding wire **110***c*, and the metal bed **120**. The sealing resin **145** has a function of protecting the MOSFET **100**, the first bonding wire **110***a*, the second bonding wire **110***b*, the third bonding wire **110***c*, and the metal bed **120**. The sealing resin **145** is, for example, an epoxy resin.
(54) For example, a source voltage is applied to the first metal lead **130***a* from the outside of the MOSFET **100**. For example, a gate voltage is applied to the second metal lead **130***b* from the outside of the MOSFET **100**. For example, a drain voltage is applied to the third metal lead **130***c* from the outside of the MOSFET **100**.
(55) FIGS. **2**A and **2**B are schematic top views of the semiconductor device according to the first embodiment. FIG. **2**A is an arrangement diagram of each region provided in the MOSFET **100**. FIG. **2**B is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET **100**.
(56) FIG. **3** is a schematic cross-sectional view of the semiconductor device according to the first embodiment. FIG. **3** is a cross-sectional view taken along line A-A′ in FIG. **2**A.
(57) FIG. **4** is a schematic cross-sectional view of the semiconductor device according to the first embodiment. FIG. **4** is a cross-sectional view taken along line B-B′ in FIG. **2**A.
(58) FIGS. **5**A and **5**B are schematic cross-sectional views of the semiconductor device according to the first embodiment. FIG. **5**A is a cross-sectional view taken along line C-C′ in FIG. **2**A. FIG. **5**B is a cross-sectional view taken along line D-D′ in FIG. **2**A.
(59) As illustrated in FIG. **2**A, the MOSFET **100** includes a transistor region **101***a* (first transistor region), a transistor region **101***b* (second transistor region), a transistor region **101***c*, a transistor region **101***d*, a diode region **102***a* (first diode region), a diode region **102***b*, and a peripheral region **103**. The transistor region **101***a* is an example of a first transistor region. The transistor region **101***b* is an example of a second transistor region. The diode region **102***a* is an example of a first diode region.
(60) Hereinafter, the transistor region **101***a*, the transistor region **101***b*, the transistor region **101***c*, and the transistor region **101***d* may be simply referred to as a transistor region **101** individually or collectively. In addition, the diode region **102***a* and the diode region **102***b* may be simply referred to as a diode region **102** individually or collectively.
(61) The MOSFET and the SBD are provided in the transistor region **101**. The SBD is provided in the diode region **102**. The MOSFET is not provided in the diode region **102**.
(62) The peripheral region **103** surrounds the transistor region **101** and the diode region **102**. In the peripheral region **103**, a gate electrode pad **22** and a gate wiring **24** are provided.
(63) In the peripheral region **103**, for example, a termination structure for improving breakdown voltage of the MOSFET **100** is provided. The termination structure for improving the breakdown voltage of the MOSFET **100** is, for example, a RESURF or a guard ring.
(64) The diode region **102** is provided between the two transistor regions **101**. For example, the diode region **102***a* is provided between the transistor region **101***a* and the transistor region **101***b*. The transistor region **101***b* is provided in the first direction parallel to a first plane P**1** with respect to the transistor region **101***a*.
(65) For example, the diode region **102***b* is provided between the transistor region **101***c* and the transistor region **101***d*. The transistor region **101***d* is provided in the first direction with respect to

the transistor region **101***c*.
(66) The width of the diode region **102** in the first direction is, for example, 30 μm or more. For example, the width of the diode region **102***a* in the first direction is 30 μm or more.
(67) The MOSFET **100** includes the silicon carbide layer **10**, the source electrode **12** (first electrode), the drain electrode **14** (second electrode), a gate insulating layer **16**, a gate electrode **18**, an interlayer insulating layer **20**, the gate electrode pad **22**, and the gate wiring **24**.
(68) The silicon carbide layer **10** includes a drain region **26** of an n.sup.+-type, a drift region **28** of an n.sup.−-type (first silicon carbide region), a body region **30** of a p-type (second silicon carbide region), a p region **32** of the p-type (fourth silicon carbide region), a source region **34** of the n.sup.+-type (third silicon carbide region), a first bottom region **36** of an n-type, and a second bottom region **38** of the n-type.
(69) The drift region **28** includes a plurality of first portions **28***a* and a plurality of second portions **28***b*. The body region **30** includes a low-concentration portion **30***a* and a high-concentration portion **30***b*. The p region **32** includes a low-concentration portion **32***a* and a high-concentration portion **32***b*.
(70) The silicon carbide layer **10** is provided between the source electrode **12** and the drain electrode **14**. The silicon carbide layer **10** is provided between the gate electrode **18** and the drain electrode **14**. The silicon carbide layer **10** is single crystal SiC. The silicon carbide layer **10** is, for example, 4H—SiC.
(71) The silicon carbide layer **10** includes a first plane ("P**1**" in FIG. **3**) and a second plane ("P**2**" in FIG. **3**). The first plane P**1** and the second plane P**2** face each other. Hereinafter, the first plane may be referred to as a surface, and the second plane may be referred to as a back surface. Hereinafter, the "depth" means a depth based on the first plane.
(72) The first plane P**1** is, for example, a face inclined at 0° or more and 8° or less with respect to the (0001) face. In addition, the second plane P**2** is, for example, a face inclined at 0° or more and 8° or less with respect to the (000-1) face. The (0001) face is referred to as a silicon face. The (000-1) face is referred to as a carbon face.
(73) The drain region **26** of the n.sup.+-type is provided on the back surface side of the silicon carbide layer **10**. The drain region **26** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the drain region **26** is, for example, 1×10.sup.1 cm.sup.−3 or more and 1×10.sup.21 cm.sup.−3 or less.
(74) The drift region **28** of the n.sup.−-type is provided between the drain region **26** and the first plane P**1**. The drift region **28** is provided between the source electrode **12** and the drain electrode **14**. The drift region **28** is provided between the gate electrode **18** and the drain electrode **14**.
(75) The drift region **28** is provided on the drain region **26**. The drift region **28** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the drift region **28** is lower than the n-type impurity concentration of the drain region **26**. The n-type impurity concentration of the drift region **28** is, for example, 4×10.sup.14 cm.sup.−3 or more and 1×10.sup.17 cm.sup.−3 or less. The thickness of the drift region **28** is, for example, 5 μm or more and 150 μm or less.
(76) The drift region **28** includes a plurality of first portions **28***a* and a plurality of second portions **28***b*. The first portion **28***a* is in contact with the first plane P**1**. The first portion **28***a* is sandwiched between the two body regions **30**. The first portion **28***a* functions as an n-type semiconductor region of the SBD. The first portion **28***a* extends, for example, in the second direction.
(77) The second portion **28***b* is in contact with the first plane P**1**. The second portion **28***b* is sandwiched between the two p regions **32**. The second portion **28***b* functions as an n-type semiconductor region of the SBD. The second portion **28***b* extends, for example, in the second direction.
(78) The body region **30** of the p-type is provided between the drift region **28** and the first plane P**1**. A part of the body region **30** functions as a channel region of the MOSFET **100**. The body region

30 functions as a p-type semiconductor region of a pn junction diode.
(79) The body region **30** includes a low-concentration portion **30***a* and a high-concentration portion **30***b*. The high-concentration portion **30***b* is provided between the low-concentration portion **30***a* and the first plane P**1**. The p-type impurity concentration of the high-concentration portion **30***b* is higher than the p-type impurity concentration of the low-concentration portion **30***a*.
(80) The body region **30** contains, for example, aluminum (Al) as a p-type impurity. The p-type impurity concentration of the low-concentration portion **30***a* is, for example, 1×10.sup.16 cm.sup.−3 or more and 5×10.sup.17 cm.sup.−3 or less. The p-type impurity concentration of the high-concentration portion **30***b* is, for example, 1×10.sup.18 cm.sup.−3 or more and 1×10.sup.21 cm.sup.−3 or less.
(81) The depth of the body region **30** is, for example, 0.3 μm or more and 1.0 μm or less.
(82) The body region **30** is fixed to the electric potential of the source electrode **12**.
(83) The p region **32** of the p-type is provided between the drift region **28** and the first plane P**1**. The p region **32** functions as a p-type semiconductor region of a pn junction diode.
(84) The p region **32** includes a low-concentration portion **32***a* and a high-concentration portion **32***b*. The high-concentration portion **32***b* is provided between the low-concentration portion **32***a* and the first plane P**1**. The p-type impurity concentration of the high-concentration portion **32***b* is higher than the p-type impurity concentration of the low-concentration portion **32***a*.
(85) The p region **32** contains, for example, aluminum (Al) as a p-type impurity. The p-type impurity concentration of the low-concentration portion **32***a* is, for example, 1×10.sup.16 cm.sup.−3 or more and 5×10.sup.17 cm.sup.−3 or less. The p-type impurity concentration of the high-concentration portion **32***b* is, for example, 1×10.sup.11 cm.sup.−3 or more and 1×10.sup.21 cm.sup.−3 or less.
(86) The p-type impurity concentration of the low-concentration portion **32***a* of the p region **32** is substantially equal to, for example, the p-type impurity concentration of the low-concentration portion **30***a* of the body region **30**.
(87) The p-type impurity concentration of the high-concentration portion **32***b* of the p region **32** is substantially equal to, for example, the p-type impurity concentration of the high-concentration portion **30***b* of the body region **30**.
(88) For example, the width of the p region **32** in the first direction is larger than the width of the body region **30** in the first direction. The depth of the p region **32** is, for example, 0.3 μm or more and 1.0 μm or less.
(89) The p region **32** is fixed to the electric potential of the source electrode **12**.
(90) The source region **34** of the n.sup.+-type is provided between the body region **30** and the first plane P**1**. The source region **34** is provided between the low-concentration portion **30***a* of the body region **30** and the first plane P**1**. The source region **34** of the n.sup.+-type extends, for example, in the second direction.
(91) The source region **34** contains, for example, phosphorus (P) as an n-type impurity. The n-type impurity concentration of the source region **34** is higher than the n-type impurity concentration of the drift region **28**.
(92) The n-type impurity concentration of the source region **34** is, for example, 1×10.sup.18 cm.sup.−3 or more and 1×10.sup.21 cm.sup.−3 or less. The depth of the source region **34** is shallower than the depth of the body region **30**. The depth of the source region **34** is, for example, 0.1 μm or more and 0.3 μm or less.
(93) The first bottom region **36** of the n-type is provided between the drift region **28** and the body region **30**. The first bottom region **36** is in contact with, for example, the drift region **28** and the body region **30**. The width of the first bottom region **36** in the first direction is, for example, substantially the same as the width of the body region **30** in the first direction.
(94) The first bottom region **36** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the first bottom region **36** is higher than the n-type impurity

concentration of the drift region **28**.
(95) The n-type impurity concentration of the first bottom region **36** is, for example, 1×10.sup.16 cm.sup.−3 or more and 2×10.sup.17 cm.sup.−3 or less. The thickness of the first bottom region **36** is, for example, 0.4 μm or more and 1.5 μm or less.
(96) The second bottom region **38** of the n-type is provided between the drift region **28** and the p region **32**. The second bottom region **38** is in contact with, for example, the drift region **28** and the p region **32**. The width of the second bottom region **38** in the first direction is, for example, substantially the same as the width of the p region **32** in the first direction.
(97) The second bottom region **38** contains, for example, nitrogen (N) as an n-type impurity. The n-type impurity concentration of the second bottom region **38** is higher than the n-type impurity concentration of the drift region **28**. The n-type impurity concentration of the second bottom region **38** is, for example, substantially the same as the n-type impurity concentration of the first bottom region **36**.
(98) The n-type impurity concentration of the second bottom region **38** is, for example, 1×10.sup.16 cm.sup.−3 or more and 2×10.sup.17 cm.sup.−3 or less. The thickness of the second bottom region **38** is, for example, 0.4 μm or more and 1.5 μm or less.
(99) The gate electrode **18** is provided on the first plane P**1** side of the silicon carbide layer **10**. The gate electrode **18** extends in the second direction parallel to the first plane P**1** and perpendicular to the first direction. A plurality of gate electrodes **18** are disposed in parallel in the first direction. The gate electrode **18** has a so-called stripe shape.
(100) The gate electrode **18** is a conductive layer. The gate electrode **18** is, for example, polycrystalline silicon containing p-type impurities or n-type impurities.
(101) The gate electrode **18** faces, for example, a portion of the body region **30** in contact with the first plane P**1**. The gate electrode **18** faces, for example, a portion of the drift region **28** in contact with the first plane P**1**.
(102) The gate insulating layer **16** is provided between the gate electrode **18** and the body region **30**. The gate insulating layer **16** is provided between the gate electrode **18** and the drift region **28**.
(103) The gate insulating layer **16** is, for example, silicon oxide. For example, a High-k insulating material (high dielectric constant insulating material) can be applied to the gate insulating layer **16**.
(104) The interlayer insulating layer **20** is provided on the gate electrode **18** and the silicon carbide layer **10**. The interlayer insulating layer **20** is provided between the gate electrode **18** and the source electrode **12**. The interlayer insulating layer **20** has a function of electrically separating the gate electrode **18** and the source electrode **12**. The interlayer insulating layer **20** is, for example, silicon oxide.
(105) The source electrode **12** is provided on the first plane P**1** side of the silicon carbide layer **10**. The source electrode **12** is in contact with the first plane P**1**.
(106) The source electrode **12** is in contact with the first portion **28***a* of the drift region **28**, the second portion **28***b* of the drift region **28**, the body region **30**, the p region **32**, and the source region **34**.
(107) The source electrode **12** has a first region **12***a* in the transistor region **101** and a second region **12***b* in the diode region **102**.
(108) The source electrode **12** contains metal. The metal forming the source electrode **12** is, for example, a stacked structure of titanium (Ti) and aluminum (Al).
(109) The portion of the source electrode **12** in contact with the body region **30**, the p region **32**, and the source region **34** is, for example, metal silicide. Metal silicide is, for example, titanium silicide or nickel silicide. For example, metal silicide is not provided in a portion of the source electrode **12** in contact with the first portion **28***a* of the drift region **28** and the second portion **28***b* of the drift region **28**.
(110) The junction between the body region **30**, the p region **32**, and the source region **34** and the source electrode **12** is, for example, an ohmic junction. The junction between the first portion **28***a*

of the drift region **28** and the second portion **28***b* of the drift region **28** and the source electrode **12** is, for example, a Schottky junction.
(111) The drain electrode **14** is provided on the second plane P**2** side of the silicon carbide layer **10**. The drain electrode **14** is in contact with the second plane P**2**. The drain electrode **14** is in contact with the drain region **26**.
(112) The drain electrode **14** is, for example, a metal or a metal semiconductor compound. The drain electrode **14** contains, for example, at least one material selected from a group formed of nickel silicide, titanium (Ti), nickel (Ni), silver (Ag), and gold (Au).
(113) The junction between the drain region **26** and the drain electrode **14** is, for example, an ohmic junction.
(114) The gate electrode pad **22** is provided on the first plane P**1** side of the silicon carbide layer **10**. The gate electrode pad **22** is provided on the interlayer insulating layer **20**. The gate electrode pad **22** is provided to implement electrical connection between the outside and the gate electrode **18**.
(115) The gate wiring **24** is provided on the first plane P**1** side of the silicon carbide layer **10**. The gate wiring **24** is connected to the gate electrode pad **22**. The gate wiring **24** is electrically connected to the gate electrode **18**.
(116) A part of the gate wiring **24** extends in the first direction parallel to the first plane P**1**. A part of the gate wiring **24** extends in the second direction parallel to the first plane P**1** and perpendicular to the first direction.
(117) The gate electrode pad **22** and the gate wiring **24** contain metal. The metal forming the gate electrode pad **22** and the gate wiring **24** is, for example, a stacked structure of titanium (Ti) and aluminum (Al). The gate electrode pad **22** and the gate wiring **24** are formed of, for example, the same metal material as the source electrode **12**.
(118) The gate wiring **24** between the two source electrodes **12** extends in the first direction. The source electrode **12** is sandwiched between the two gate wirings **24** extending in the first direction. The source electrode **12** is sandwiched between the two gate wirings **24** extending in the second direction.
(119) As illustrated in FIG. **2**B, a connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is provided across the first region **12***a* and the second region **12***b*. The connection portion **110***ax* is in contact with both the first region **12***a* and the second region **12***b*.
(120) As illustrated in FIG. **3**, the transistor region **101** includes the silicon carbide layer **10**, the source electrode **12** (first electrode), the drain electrode **14** (second electrode), the gate insulating layer **16**, the gate electrode **18**, and the interlayer insulating layer **20**. The silicon carbide layer **10** of the transistor region **101** includes the drain region **26** of the n.sup.+-type, the drift region **28** of the n.sup.−-type (first silicon carbide region), the body region **30** of the p-type (second silicon carbide region), the source region **34** of the n.sup.+-type (third silicon carbide region), and the first bottom region **36** of the n-type. The drift region **28** of the transistor region **101** includes the plurality of first portions **28***a*.
(121) In the transistor region **101**, the source electrode **12**, the first portion **28***a* of the drift region **28**, the drain region **26**, and the drain electrode **14** form the SBD. The source electrode **12**, the body region **30**, the first bottom region **36**, the drain region **26**, and the drain electrode **14** form the pn junction diode.
(122) A first distance (d**1** in FIGS. **5**A and **5**B) between the two first portions **28***a* adjacent to each other with the body region **30** interposed therebetween is, for example, 3 μm or more and 30 μm or less.
(123) As illustrated in FIG. **4**, the diode region **102** includes the silicon carbide layer **10**, the source electrode **12** (first electrode), and the drain electrode **14** (second electrode). The silicon carbide layer **10** of the diode region **102** includes the drain region **26** of the n.sup.+-type, the drift region **28** of the n.sup.−-type (first silicon carbide region), the p region **32** of the p-type (fourth silicon

carbide region), and the second bottom region **38** of the n-type. The drift region **28** of the diode region **102** includes the plurality of second portions **28***b*.
(124) In the diode region **102**, the source electrode **12**, the second portion **28***b* of the drift region **28**, the drain region **26**, and the drain electrode **14** form the SBD. The source electrode **12**, the p region **32**, the second bottom region **38**, the drain region **26**, and the drain electrode **14** form the pn junction diode.
(125) A second distance (d**2** in FIGS. **5**A and **5**B) between the two second portions **28***b* adjacent to each other with the p region **32** interposed therebetween is, for example, 3 μm or more and 30 μm or less. The second distance d**2** between the two second portions **28***b* adjacent to each other with the p region **32** interposed therebetween is, for example, substantially equal to the first distance d**1** between the two first portions **28***a* adjacent to each other with the body region **30** interposed therebetween. The first distance d**1** and the second distance d**2** are distances in the first direction.
(126) FIG. **6** is a schematic top view of the semiconductor device according to the first embodiment. FIG. **6** is a view illustrating a pattern of the body region **30** projected onto the first plane P**1** and a pattern of the p region **32** projected onto the first plane P**1**. The pattern of the body region **30** and the pattern of the p region **32** in FIG. **6** are patterns projected onto the first plane P**1** in a direction perpendicular to the first plane P**1**.
(127) An occupancy rate per unit area of the p region **32** projected onto the first plane P**1** on the first plane P**1** is larger than an occupancy rate per unit area of the body region **30** projected onto the first plane P**1** on the first plane P**1**. In other words, in the region of a predetermined size, the occupancy rate of the p region **32** projected onto the first plane P**1** on the first plane P**1** is larger than the occupancy rate of the body region **30** projected onto the first plane P**1** on the first plane P**1**. The occupancy rate is an occupancy rate with respect to the transistor region **101** and the diode region **102** projected onto the first plane P**1**. That is, the occupancy ratio of the pn junction diode in the diode region **102** is larger than the occupancy ratio of the pn junction diode in the transistor region **101**.
(128) The occupancy rate per unit area of the p region **32** projected onto the first plane P**1** is, for example, 1.2 times or more and 3 times or less the occupancy rate per unit area of the body region **30** projected onto the first plane P**1**.
(129) The unit area is not particularly limited as long as the average occupancy rate of the body region **30** of the transistor region **101** with the average occupancy rate of the p region **32** of the diode region **102** can be compared. The unit area is, for example, 30 μm×30 μm=900 μm.sup.2.
(130) In addition, a contact area per unit area between the source electrode **12** and the p region **32** in the diode region **102** is larger than a contact area per unit area between the source electrode **12** and the body region **30** in the transistor region **101**. That is, contact resistance per unit area between the source electrode **12** and the p region **32** in the diode region **102** is smaller than contact resistance per unit area between the source electrode **12** and the body region **30** in the transistor region **101**.
(131) FIG. **7** is a schematic top view of the semiconductor device according to the first embodiment. FIG. **7** illustrates a shape of a contact plane CP**1** between the first bonding wire **110***a* and the source electrode **12**.
(132) As illustrated in FIG. **2**B, a connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is provided across the first region **12***a* and the second region **12***b*. The connection portion **110***ax* is in contact with both the first region **12***a* and the second region **12***b*. Therefore, the contact plane CP**1** between the first bonding wire **110***a* and the source electrode **12** extends between the first region **12***a* and the second region **12***b*.
(133) A first contact area S**1** between the connection portion **110***ax* and the first region **12***a* is larger than a second contact area S**2** between the connection portion **110***ax* and the second region **12***b*. In other words, the area of the contact plane CP**1** on the first region **12***a* is larger than the area of the contact plane CP**1** on the second region **12***b*.

(134) The first contact area S**1** is, for example, four times or more the second contact area S**2**. The first direction (w**1** in FIG. **7**) of the contact plane CP**1** is, for example, 1 mm or more.
(135) Next, functions and effects of the semiconductor device according to the first embodiment will be described.
(136) FIGS. **8**A and **8**B are schematic top views of a semiconductor device according to a first comparative example. FIG. **8**A is an arrangement diagram of each region included in an MOSFET **901** of the first comparative example. FIG. **8**B is a diagram showing a pattern of an electrode and a wiring on the upper face of the MOSFET **901** of the first comparative example. FIGS. **8**A and **8**B are views corresponding to FIGS. **2**A and **2**B of the first embodiment.
(137) FIG. **9** is a schematic cross-sectional view of the semiconductor device according to the first comparative example. FIG. **9** is a cross-sectional view taken along line G-G′ in FIG. **8**A. FIG. **9** is a diagram corresponding to FIG. **5**A of the first embodiment.
(138) The MOSFET **901** of the first comparative example is different from the MOSFET **100** of the first embodiment in that the diode region **102** is not provided.
(139) In the transistor region **101** of the MOSFET **901** of the first comparative example, the MOSFET and the SBD are provided in the same manner as that of the MOSFET **100** of the first embodiment.
(140) FIG. **10** is an equivalent circuit diagram of the semiconductor device according to the first comparative example. Between the source electrode **12** and the drain electrode **14**, the pn junction diode and the SBD are connected as built-in diodes in parallel with the transistor.
(141) For example, a case where the MOSFET is used as a switching element connected to an inductive load will be considered. When the MOSFET **901** is turned off, a voltage at which the source electrode **12** is positive with respect to the drain electrode **14** may be applied due to a load current caused by an inductive load. In this case, a forward current flows through the built-in diode. This state is also referred to as a reverse conduction state.
(142) A forward voltage (Vf) at which a forward current starts flowing through the SBD is lower than a forward voltage (Vf) of the pn junction diode. Therefore, first, the forward current flows through the SBD.
(143) The forward voltage (Vf) of the SBD is, for example, 1.0 V. The forward voltage (Vf) of the pn junction diode is, for example, 2.5 V.
(144) The SBD performs unipolar operation. Therefore, even if a forward current flows, a stacking fault does not grow in the silicon carbide layer **10** due to recombination energy of carriers.
(145) FIGS. **11**A and **11**B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment. FIGS. **11**A and **11**B are schematic cross-sectional views of the semiconductor device according to the first comparative example. FIGS. **11**A and **11**B are views corresponding to FIG. **9**.
(146) FIGS. **11**A and **11**B are diagrams illustrating a current flowing through the built-in diode of the MOSFET **901** of the first comparative example. FIG. **11**A illustrates a state in which a forward current flows only through the SBD, and FIG. **11**B illustrates a state in which a forward current flows through the SBD and the pn junction diode.
(147) That is, FIG. **11**A illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is lower than the forward voltage (Vf) of the pn junction diode. FIG. **11**B illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (Vf) of the pn junction diode.
(148) In FIGS. **11**A and **11**B, a dotted arrow indicates a current flowing through the SBD. In FIG. **11**B, a solid arrow indicates a current flowing through the pn junction diode.
(149) As illustrated in FIG. **11**A, the current flowing through the SBD flows around the bottom portion of the body region **30**. Therefore, the electrostatic potential wraps around in the drift region **28** facing the bottom portion of the body region **30**. The wraparound of the electrostatic potential reduces the voltage applied between the body region **30** and the drift region **28**.

(150) Therefore, at the bottom of the body region **30**, the voltage hardly exceeds the forward voltage (Vf) of the pn junction diode. In other words, the forward voltage (Vf) of the pn junction diode of the MOSFET **901** of the first comparative example can be made higher than a case where the SBD is not provided. Therefore, bipolar operation of the pn junction diode is suppressed, and formation of a stacking fault in the silicon carbide layer **10** due to recombination energy of carriers is suppressed.
(151) The forward voltage (Vf) of the pn junction diode of the MOSFET **901** of the first comparative example depends on a distance between two SBDs adjacent to each other in the first direction. By reducing the distance between the two SBDs adjacent to each other in the first direction, the forward voltage (Vf) of the pn junction diode of the MOSFET **901** of the first comparative example can be increased.
(152) A large surge current exceeding a steady state may be instantaneously applied to the MOSFET. The surge current flows from the source electrode **12** toward the drain electrode **14**.
(153) When a large surge current flows, a large surge voltage is applied, and the MOSFET generates heat, which causes a breakdown of the MOSFET. The maximum allowable peak current value (I.sub.FSM) of the surge current allowed in the MOSFET is referred to as a surge current tolerance. In the MOSFET having the SBD provided therein, it is desired to improve a surge current tolerance.
(154) When a large surge voltage is applied to the MOSFET **901** of the first comparative example, the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (Vf) of the pn junction diode.
(155) When the voltage applied between the pn junctions of the pn junction diode becomes higher than the forward voltage (Vf) of the pn junction diode, a current also flows through the pn junction diode, as illustrated in FIG. **11**B.
(156) FIG. **12** is a schematic cross-sectional view of a semiconductor device according to a second comparative example. FIG. **12** is a diagram corresponding to FIG. **9** of the first comparative example.
(157) An MOSFET **902** of the second comparative example is different from the MOSFET **901** of the first comparative example in that the transistor region does not include the SBD. The built-in diode of the MOSFET **902** of the second comparative example is only the pn junction diode.
(158) FIG. **13** is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment. FIG. **13** is a diagram illustrating voltage-current characteristics of the built-in diodes of the MOSFET **901** of the first comparative example and the MOSFET **902** of the second comparative example.
(159) As illustrated in FIG. **13**, in the MOSFET **902** of the second comparative example, when a voltage equal to or higher than a forward voltage Vf**2** of the pn junction diode is applied, a current flows through the pn junction diode. On the other hand, in the MOSFET **901** of the first comparative example, a current flows through the SBD until a forward voltage Vf**1** of the pn junction diode is applied. In the MOSFET **901** of the first comparative example, when a voltage equal to or higher than the forward voltage Vf**1** of the pn junction diode is applied, a current flows through the pn junction diode.
(160) Since the MOSFET **901** of the first comparative example performs the unipolar operation up to the forward voltage Vf**1**, the slope of the current increase is smaller than that of the MOSFET **902** of the second comparative example. Therefore, a maximum allowable peak current value I.sub.FSM**1** of the MOSFET **901** of the first comparative example is smaller than a maximum allowable peak current value I.sub.FSM**2** of the MOSFET **902** of the second comparative example. In other words, a surge current tolerance of the MOSFET **901** of the first comparative example is smaller than a surge current tolerance of the MOSFET **902** of the second comparative example.
(161) An MOSFET **903** of a third comparative example is different from the MOSFET **100** of the first embodiment only in that the connection portion **110***ax* of the first bonding wire **110***a* to the

source electrode **12** is not in contact with the first region **12***a* but is in contact only with the second region **12***b*.
(162) The MOSFET **903** of the third comparative example includes the diode region **102** provided between the transistor regions **101** in the same manner as that of the MOSFET **100** of the first embodiment. Since the MOSFET **903** of the third comparative example includes the diode region **102**, a surge current tolerance is improved. Hereinafter, a description will be given in detail.
(163) FIGS. **14**A and **14**B are explanatory diagrams of functions and effects of the semiconductor device according to the first embodiment. FIGS. **14**A and **14**B are schematic cross-sectional views of the MOSFET **903** of the third comparative example. FIGS. **14**A and **14**B are views corresponding to FIG. **5**A.
(164) FIGS. **14**A and **14**B are diagrams illustrating a current flowing through the built-in diode of the MOSFET **903** of the third comparative example. FIG. **14**A illustrates a state in which a forward current flows only through the SBD, and FIG. **14**B illustrates a state in which a forward current flows through the SBD and the pn junction diode.
(165) That is, FIG. **14**A illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is lower than the forward voltage (Vf) of the pn junction diode. FIG. **14**B illustrates a state in which the voltage applied between the pn junctions of the pn junction diode is higher than the forward voltage (Vf) of the pn junction diode.
(166) In FIGS. **14**A and **14**B, a dotted arrow indicates a current flowing through the SBD. In FIG. **14**B, a solid arrow indicates a current flowing through the pn junction diode.
(167) The second distance d**2** between the two second portions **28***b* adjacent to each other with the p region **32** interposed therebetween in the diode region **102** is substantially equal to the first distance d**1** between the two first portions **28***a* adjacent to each other with the body region **30** interposed therebetween in the transistor region **101**. In other words, the diode region **102** is provided with the second portion **28***b* at the same distance as the first portion **28***a* of the transistor region **101**. In other words, an SBD region is provided in the diode region **102** at the same distance as the transistor region **101**.
(168) Therefore, as illustrated in FIG. **14**A, in the diode region **102**, the current flowing through the SBD flows around the bottom of the p region **32**. Therefore, at the bottom portion of the p region **32**, the voltage hardly exceeds the forward voltage (Vf) of the pn junction diode. The forward voltage (Vf) of the pn junction diode in the diode region **102** is increased by providing the SBD region.
(169) When the voltage applied between the pn junctions of the pn junction diode becomes higher than the forward voltage (Vf) of the pn junction diode, a current also flows through the pn junction diode, as illustrated in FIG. **14**B.
(170) In the MOSFET **903** of the third comparative example, the occupancy rate per unit area of the p region **32** projected onto the first plane P**1** is larger than the occupancy rate per unit area of the body region **30** projected onto the first plane P**1**. That is, the occupancy ratio of the pn junction diode in the diode region **102** is larger than the occupancy ratio of the pn junction diode in the transistor region **101**.
(171) In addition, a contact area per unit area between the source electrode **12** and the p region **32** in the diode region **102** is larger than a contact area per unit area between the source electrode **12** and the body region **30** in the transistor region **101**. That is, contact resistance per unit area between the source electrode **12** and the p region **32** in the diode region **102** is smaller than contact resistance per unit area between the source electrode **12** and the body region **30** in the transistor region **101**.
(172) Therefore, the current flowing through the pn junction diode of the diode region **102** is larger than the current flowing through the pn junction diode of the transistor region **101**.
(173) In addition, when a large current flows through the pn junction diode of the diode region **102**, carrier propagation and heat propagation to the adjacent transistor region **101** occur. Therefore,

conductivity modulation of the transistor region **101** adjacent to the diode region **102** is promoted. Therefore, the current flowing through the pn junction diode of the transistor region **101** adjacent to the diode region **102** increases.
(174) FIG. **15** is an explanatory diagram of functions and effects of the semiconductor device according to the first embodiment. FIG. **15** is a diagram illustrating voltage-current characteristics of the built-in diodes of the MOSFET **901** of the first comparative example, the MOSFET **902** of the second comparative example, and the MOSFET **903** of the third comparative example.
(175) As illustrated in FIG. **15**, in the MOSFET **903** of the third comparative example, a current flows through the SBD until the forward voltage Vf**3** of the pn junction diode is applied. In the MOSFET **903** of the third comparative example, when a voltage equal to or higher than the forward voltage Vf**3** of the pn junction diode is applied, a current flows through the pn junction diode.
(176) In the diode region **102** of the MOSFET **903** of the third comparative example, the SBD region is provided at the same distance as the transistor region **101**. Therefore, the forward voltage Vf**3** of the pn junction diode of the MOSFET **903** of the third comparative example is equal to the forward voltage Vf**1** of the pn junction diode of the MOSFET of the first comparative example.
(177) Meanwhile, a current after exceeding the forward voltage Vf**3** of the pn junction diode in the MOSFET **903** of the third comparative example is larger than a current after exceeding the forward voltage Vf**1** of the pn junction diode in the MOSFET **901** of the first comparative example. This is because the current flowing through the pn junction diode of the diode region **102** and the pn junction diode of the transistor region **101** adjacent to the diode region **102** is larger than that of the MOSFET **901** of the first comparative example.
(178) Since the current after exceeding the forward voltage Vf**3** of the pn junction diode increases, a maximum allowable peak current value I.sub.FSM**3** of the MOSFET **903** of the third comparative example becomes larger than the maximum allowable peak current value I.sub.FSM**1** of the MOSFET **901** of the first comparative example. In other words, the surge current tolerance of the MOSFET **903** of the third comparative example is larger than the surge current tolerance of the MOSFET **901** of the first comparative example.
(179) As described above, the MOSFET **903** of the third comparative example includes the diode region **102** provided between the transistor regions **101**, thereby improving the surge current tolerance.
(180) The occupancy rate per unit area of the p region **32** projected onto the first plane P**1** is preferably 1.2 times or more and 3 times or less the occupancy rate per unit area of the body region **30** projected onto the first plane P**1**. By exceeding the lower limit value, the surge current tolerance is further improved. In addition, by falling below the upper limit value, a decrease in the forward voltage Vf**3** is suppressed, and a decrease in reliability is suppressed.
(181) FIGS. **16**A and **16**B are schematic top views of a semiconductor device according to the third comparative example. FIG. **16**A is an arrangement diagram of each region provided in the MOSFET **903**. FIG. **16**B is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET **903**.
(182) As illustrated in FIG. **16**B, the connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is not in contact with the first region **12***a* but is in contact only with the second region **12***b*.
(183) FIG. **17** is a schematic top view of the semiconductor device according to the third comparative example. FIG. **17** illustrates a shape of a contact plane CP**2** between the first bonding wire **110***a* and the source electrode **12**. FIG. **17** is a diagram corresponding to FIG. **7** of the first embodiment.
(184) The contact plane CP**2** between the first bonding wire **110***a* and the source electrode **12** exists only in the second region **12***b*. The contact plane CP**2** between the first bonding wire **110***a* and the source electrode **12** does not exist in the first region **12***a*.

(185) The first contact area S**1** between the connection portion **110***ax* and the first region **12***a* is zero. Therefore, the first contact area S**1** between the connection portion **110***ax* and the first region **12***a* is smaller than the second contact area S**2** between the connection portion **110***ax* and the second region **12***b*. In other words, the area of the contact plane CP**2** on the first region **12***a* is smaller than the area of the contact plane CP**2** on the second region **12***b*.
(186) FIGS. **18**A and **18**B are explanatory diagrams of a problem of the semiconductor device according to the third comparative example. FIGS. **18**A and **18**B are views corresponding to FIGS. **2**A and **2**B of the first embodiment.
(187) As described above, the MOSFET **903** of the third comparative example includes the diode region **102** provided between the transistor regions **101**, thereby improving the surge current tolerance. Meanwhile, in the MOSFET **903** of the third comparative example, there is a problem in that variations in surge current tolerance between chips are large. In other words, there is a problem in that a chip having a low surge current tolerance may be accidentally generated.
(188) As a result of failure analysis of a chip having a low surge current tolerance by the inventors, it has become clear that one of the causes of the low surge current tolerance is a short circuit between the source electrode **12** and the gate wiring **24**. It has become clear that the short circuit between the source electrode **12** and the gate wiring **24** occurs when the source electrode **12** adjacent to the gate wiring **24** melts and flows in the lateral direction to come into contact with the gate wiring **24**, as indicated by a dotted circle in FIG. **18**B.
(189) It has become clear that a short circuit between the source electrode **12** and the gate wiring **24** is likely to occur in the second direction of the connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12**, as shown in FIG. **18**B.
(190) It is considered that, in the MOSFET **903** of the third comparative example, the short circuit between the source electrode **12** and the gate wiring **24** is likely to occur in the second direction of the connection portion **110***ax* of the first bonding wire **110***a* because the connection portion **110***ax* is in contact only with the second region **12***b* of the diode region **102**.
(191) When calorific values between the transistor region **101** and the diode region **102** are compared with each other, the surge current flowing through the diode region **102** is larger, so the calorific value is larger. In addition, the current flowing through the first bonding wire **110***a* also causes the connection portion **110***ax* of the first bonding wire **110***a* to generate heat.
(192) Since the connection portion **110***ax* is in contact only with the second region **12***b*, interaction between the heat generation of the diode region **102** and the heat generation of the connection portion **110***ax* of the first bonding wire **110***a* increases, and as such, the heat generation of the second region **12***b* increases. Therefore, it is considered that the second region **12***b* is likely to melt. It is considered that the second region **12***b* to which the first bonding wire **110***a* is connected melts, causing the short circuit between the source electrode **12** and the gate wiring **24**.
(193) In the MOSFET **100** of the first embodiment, as illustrated in FIG. **2**B, the connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is provided across the first region **12***a* and the second region **12***b*. As illustrated in FIG. **7**, the first contact area S**1** between the connection portion **110***ax* and the first region **12***a* is larger than the second contact area S**2** between the connection portion **110***ax* and the second region **12***b*.
(194) Therefore, as compared with the MOSFET **903** of the third comparative example, interaction between the heat generation of the diode region **102** and the heat generation of the connection portion **110***ax* of the first bonding wire **110***a* is reduced. Therefore, melting of the source electrode **12** is suppressed, and the short circuit between the source electrode **12** and the gate wiring **24** is suppressed.
(195) From the viewpoint of suppressing the short circuit between the source electrode **12** and the gate wiring **24**, the first contact area S**1** is preferably 4 times or more, more preferably 10 times or more the second contact area S**2**.
(196) As described above, according to the first embodiment, a discrete device with an improved

surge current tolerance is realized.

Second Embodiment

(197) A semiconductor device according to a second embodiment is different from the semiconductor device according to the first embodiment in that a conductor is not in contact with a second region. Hereinafter, some descriptions of contents overlapping with the first embodiment may be omitted.

(198) FIGS. **19**A and **19**B are schematic top views of the semiconductor device according to the second embodiment. FIG. **19**A is an arrangement diagram of each region provided in an MOSFET **200** of the second embodiment. FIG. **19**B is a diagram illustrating a pattern of an electrode and a wiring on the upper face of the MOSFET **200**. FIGS. **19**A and **19**B are views corresponding to FIGS. **2**A and **2**B of the first embodiment.

(199) As illustrated in FIG. **19**B, the connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is not in contact with the second region **12***b* but is in contact only with the first region **12***a*.

(200) FIG. **20** is a schematic top view of the semiconductor device according to the second embodiment. FIG. **20** illustrates a shape of a contact plane CP**3** between the first bonding wire **110***a* and the source electrode **12**. FIG. **20** is a diagram corresponding to FIG. **7** of the first embodiment.

(201) The contact plane CP**3** between the first bonding wire **110***a* and the source electrode **12** exists only in the first region **12***a*. The contact plane CP**3** between the first bonding wire **110***a* and the source electrode **12** does not exist in the second region **12***b*.

(202) The second contact area S**2** between the connection portion **110***ax* and the second region **12***b* is zero. Therefore, the first contact area S**1** between the connection portion **110***ax* and the first region **12***a* is larger than the second contact area S**2** between the connection portion **110***ax* and the second region **12***b*. In other words, the area of the contact plane CP**3** on the first region **12***a* is larger than the area of the contact plane CP**3** on the second region **12***b*.

(203) In the MOSFET **200** of the second embodiment, the connection portion **110***ax* of the first bonding wire **110***a* to the source electrode **12** is not in contact with the second region **12***b*. Therefore, as compared with the MOSFET **100** of the first embodiment, interaction between the heat generation of the diode region **102** and the heat generation of the connection portion **110***ax* of the first bonding wire **110***a* is further reduced. Therefore, melting of the source electrode **12** is suppressed, and the short circuit between the source electrode **12** and the gate wiring **24** is further suppressed.

(204) It is noted that, from the viewpoint of not bringing the connection portion **110***ax* into contact with the second region **12***b*, the width of the first transistor region **101***a* in the first direction is preferably larger than the width of the contact plane CP**3** in the first direction (w**3** in FIG. **20**).

(205) As described above, according to the second embodiment, a discrete device with an improved surge current tolerance is realized.

Third Embodiment

(206) An inverter circuit and a driving device according to a third embodiment are an inverter circuit and a driving device including the semiconductor device according to the first embodiment.

(207) FIG. **21** is a schematic diagram of the driving device according to the third embodiment. A driving device **500** includes a motor **140** and an inverter circuit **150**.

(208) The inverter circuit **150** includes three semiconductor modules **150***a*, **150***b*, and **150***c* using the MOSFET **100** of the first embodiment as a switching element. By connecting the three semiconductor modules **150***a*, **150***b*, and **150***c* in parallel, a three-phase inverter circuit **150** including three AC voltage output terminals U, V, and W is implemented. The motor **140** is driven by the AC voltage output from the inverter circuit **150**.

(209) According to the third embodiment, the characteristics of the inverter circuit **150** and the driving device **500** are improved by including the MOSFET **100** with improved characteristics.

Fourth Embodiment

(210) A vehicle according to a fourth embodiment is a vehicle including the semiconductor device according to the first embodiment.

(211) FIG. **22** is a schematic diagram of the vehicle according to the fourth embodiment. A vehicle **600** according to the fourth embodiment is a railway vehicle. The vehicle **900** includes a motor **140** and an inverter circuit **150**.

(212) The inverter circuit **150** includes three semiconductor modules using the MOSFET **100** of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit **150** including three AC voltage output terminals U, V, and W is implemented. The motor **140** is driven by the AC voltage output from the inverter circuit **150**. Wheels **90** of the vehicle **600** are rotated by the motor **140**.

(213) According to the fourth embodiment, characteristics of the vehicle **600** are improved by providing the MOSFET **100** with improved characteristics.

Fifth Embodiment

(214) A vehicle according to a fifth embodiment is a vehicle including the semiconductor device according to the first embodiment.

(215) FIG. **23** is a schematic diagram of the vehicle according to the fifth embodiment. A vehicle **700** according to the fifth embodiment is an automobile. The vehicle **700** includes a motor **140** and an inverter circuit **150**.

(216) The inverter circuit **150** includes three semiconductor modules using the MOSFET **100** of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit **150** including three AC voltage output terminals U, V, and W is implemented.

(217) The motor **140** is driven by the AC voltage output from the inverter circuit **150**. Wheels **90** of the vehicle **700** are rotated by the motor **140**.

(218) According to the fifth embodiment, characteristics of the vehicle **700** are improved by providing the MOSFET **100** with improved characteristics.

Sixth Embodiment

(219) An elevator according to a sixth embodiment is an elevator including the semiconductor device according to the first embodiment.

(220) FIG. **24** is a schematic diagram of an elevator (lift) according to the sixth embodiment. An elevator **800** according to the sixth embodiment includes a car **610**, a counterweight **612**, a wire rope **614**, a hoisting machine **616**, a motor **140**, and an inverter circuit **150**.

(221) The inverter circuit **150** includes three semiconductor modules using the MOSFET **100** of the first embodiment as a switching element. By connecting the three semiconductor modules in parallel, the three-phase inverter circuit **150** including three AC voltage output terminals U, V, and W is implemented.

(222) The motor **140** is driven by the AC voltage output from the inverter circuit **150**. The hoisting machine **616** is rotated by the motor **140**, and the car **610** moves upwards and downwards.

(223) According to the sixth embodiment, the characteristics of the elevator **800** are improved by providing the MOSFET **100** with improved characteristics.

(224) In the first or second embodiment, the case of 4H—SiC has been described as an example of a crystal structure of SiC, but the present disclosure can also be applied to devices using SiC having other crystal structures such as 6H—SiC and 3C—SiC. Further, it is also possible to apply a face other than the (0001) face to the surface of the silicon carbide layer **10**.

(225) In the first or second embodiment, the case where the gate electrode **18** has a so-called stripe shape has been described as an example, but the shape of the gate electrode **18** is not limited to the stripe shape. For example, the shape of the gate electrode **18** may be a lattice shape.

(226) In the first or second embodiment, aluminum (Al) is exemplified as the p-type impurity, but boron (B) can also be used. In addition, although nitrogen (N) and phosphorus (P) have been

exemplified as the n-type impurity, arsenic (As), antimony (Sb), and the like can also be applied.
(227) In the first and second embodiments, the case where the conductor is the bonding wire has been described as an example, but the conductor may be, for example, a clip used for clip bonding.
(228) In the first and second embodiments, the case where a semiconductor device is a discrete device has been described as an example, but the semiconductor device may be a module device on which a plurality of semiconductor chips are mounted.
(229) The number of the transistor regions **101** and the diode regions **102** and the arrangement of the transistor regions **101** and the diode regions **102** are not limited to those of the first and second embodiments.
(230) In addition, in the third to sixth embodiments, the configuration including the MOSFET **100** of the first embodiment has been described as an example, but a configuration including the MOSFET of the second embodiment is also possible.
(231) In the third to sixth embodiments, the case where the semiconductor device of the present disclosure is applied to a vehicle or an elevator has been described as an example, but the semiconductor device of the present disclosure can also be applied to, for example, a power conditioner of a solar power generation system.
(232) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the semiconductor device, the inverter circuit, the driving device, the vehicle, and the elevator described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the devices and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A semiconductor device comprising: a semiconductor chip including a plurality of transistor regions and at least one diode region, wherein the transistor regions include: a silicon carbide layer having a first plane and a second plane facing the first plane, the silicon carbide layer including a first silicon carbide region of n-type having a plurality of first portions in contact with the first plane, a second silicon carbide region of p-type provided between the first silicon carbide region and the first plane, and a third silicon carbide region of n-type provided between the second silicon carbide region and the first plane; a first electrode in contact with the first portions, the second silicon carbide region, and the third silicon carbide region; a second electrode in contact with the second plane; a gate electrode facing the second silicon carbide region; and a gate insulating layer provided between the gate electrode and the second silicon carbide region, wherein the at least one diode region includes: the silicon carbide layer including the first silicon carbide region of n-type having a plurality of second portions in contact with the first plane and a fourth silicon carbide region of p-type provided between the first silicon carbide region and the first plane; the first electrode in contact with the second portions and the fourth silicon carbide region; and the second electrode, wherein an occupied area per unit area of the fourth silicon carbide region projected onto the first plane is larger than an occupied area per the unit area of the second silicon carbide region projected onto the first plane, and wherein a first diode region being one of the at least one diode region is provided between a first transistor region and a second transistor region, the first transistor region is one of the transistor regions, the second transistor region is one of the transistor regions provided in a first direction with respect to the first transistor region; and a conductor having one end in contact with the first electrode and configured to apply a voltage to the first electrode, wherein the first electrode includes a first region in the transistor regions and a second region in the at least one diode region, and wherein a first contact area between the conductor and the first region

is larger than a second contact area between the conductor and the second region.

2. The semiconductor device according to claim 1, further comprising: a first metal layer facing the second electrode, the first metal layer being electrically connected to the second electrode; and a second metal layer in contact with the other end of the conductor.

3. The semiconductor device according to claim 1, wherein the first contact area is four times or more the second contact area.

4. The semiconductor device according to claim 1, wherein the conductor is not in contact with the second region.

5. The semiconductor device according to claim 1, wherein the conductor is a bonding wire.

6. The semiconductor device according to claim 1, wherein a contact area per the unit area between the first electrode and the fourth silicon carbide region is larger than a contact area per the unit area between the first electrode and the second silicon carbide region.

7. The semiconductor device according to claim 1, wherein a second distance between two of the second portions adjacent to each other with the fourth silicon carbide region interposed therebetween is equal to a first distance between two of the first portions adjacent to each other with the second silicon carbide region interposed therebetween.

8. The semiconductor device according to claim 1, wherein a width of the first diode region in the first direction is 30 μm or more.

9. The semiconductor device according to claim 1, wherein a width of the transistor region in the first direction is larger than a width of a contact plane between the conductor and the first electrode in the first direction.

10. An inverter circuit comprising the semiconductor device according to claim 1.

11. A driving device comprising the semiconductor device according to claim 1.

12. A vehicle comprising the semiconductor device according to claim 1.

13. An elevator comprising the semiconductor device according to claim 1.